

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1610

Rhome 138 kV
August 29, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1610. The Interconnection Customer proposes a 66.3 MW battery energy storage facility interconnecting at the Rhome 138 kV substation in the Pennsylvania area. The request was received on November 18, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 430 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,485,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of S Beach / Greenback-Fairoaks 138 kV, the S Beach / Rhome 138 kV L loads to 148.0% of its 186 MVA rating. Project DFAX: 43.2%.

Cartana / Cuthbert 138 kV: 128.8% loading (350 MVA) following the outage of the Guilbeau Electric 345-138 kV Autotransformer; DFAX = 34.4%.

The Guilbeau Electric 345 kV/138 kV Transformer Bank is overloaded to 126.5% of its 443 MVA emergency rating for the loss of the Liberty Electric Power Plant Substation / Lake Catherine 138 kV. The project contributes 10.1% DFAX (6.7 MW) to this overload.

Wrightwood Substation / Mink Street 138 kV: 126.1% loading (176 MVA) following the outage of the Guilbeau Electric 345/138 kV Transformer; DFAX = 5.7%.

The Woodlynne / Rhome 138 kV line is overloaded to 125.6% of its 207 MVA emergency rating for the loss of the S Beach / Greenback-Fairoaks 138 kV L. The project contributes 56.8% DFAX (37.7 MW) to this overload.

For the loss of Guilbeau Electric 345/138kV Transformer, the South Beach / Cuhbert 138 kV loads to 124.7% of its 339 MVA rating. Project DFAX: 34.4%.

For the loss of Guilbea Electric / Oak Hall 138 kV, the Mount Holly / Guilbeau Electric 138 kV loads to 116.8% of its 366 MVA rating. Project DFAX: 7.4%.

For the loss of Woodlynne / Rhome 138 kV, the S Beach / Greenback-Fairoaks 138 kV L loads to 116.1% of its 223 MVA rating. Project DFAX: 55.8%.

Msc / Cartanza 138 kV Cir. 2: 114.7% loading (297 MVA) following the outage of the Cartanza Substation / Msc 138 kV Ckt 1; DFAX = 16.3%.

Castleton Substation / Plainfield 138 kV L: 113.7% loading (337 MVA) following the outage of the Marcus Hook Energy Lp Substation / Bridgeport 138 kV L; DFAX = 5.2%.

The Trainer / Bridgeport 138 kV is overloaded to 110.1% of its 331 MVA emergency rating for the loss of the Bridgeprt / Marcus Hook Energy Lp 138 kV L. The project contributes 5.2% DFAX (3.5 MW) to this overload.

The Greenback-Fairoaks Substation / La Parguera 6603 138 kV is overloaded to 108.3% of its 192 MVA emergency rating for the loss of the Rhome / Woodlynne 138 kV. The project contributes 55.8% DFAX (37.0 MW) to this overload.

The Msc / Caranza Substation 138 kV #1 is overloaded to 107.9% of its 315 MVA emergency rating for the loss of the Msc / Cartanza 138 kV L. The project contributes 16.3% DFAX (10.8 MW) to this overload.

SuperMax / Chase 138 kV: 105.1% loading (418 MVA) following the outage of the Chilliick / 2Th And F 138 kV L; DFAX = 6.7%.

Super Max / Gren Country 138 kV: 104.7% loading (432 MVA) following the outage of the 26h And F / Chilliwick 138 kV line; DFAX = 6.7%.

For the loss of South Beach / Greenback-Fairoaks 138 kV L, the Hurricane Crk Substation / Woodlynne 138 kV loads to 102.2% of its 160 MVA rating. Project DFAX: 37.0%.

The St. Johns / Fruitland 138 kV is overloaded to 101.7% of its 233 MVA emergency rating for the loss of the Jarrett / Heaton 138 kV. The project contributes 5.6% DFAX (3.7 MW) to this overload.

For the loss of Jarrett / Heaton 138 kV line, the St. Johns / Hathaway Street 138 kV loads to 100.3% of its 273 MVA rating. Project DFAX: 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
South Beach / Rhome 138 kV	Rebuild line with higher-capacity conductor	\$409,000	100% (\$409,000)
Cartana / Cuthbert 138 kV	Rebuild line with higher-capacity conductor	\$9,134,000	100% (\$9,134,000)
Guilbeau Electric 345/138kV Auto	Replace transformer with higher-rated unit	\$13,065,000	100% (\$13,065,000)
Wrightwood / Mink Street 138 kV	Rebuild line with higher-capacity conductor	\$3,292,000	100% (\$3,292,000)
Woodlynne / Rhome 138 kV line	Rebuild line with higher-capacity conductor	\$1,026,000	100% (\$1,026,000)
S Beach / Cuthbert 138 kV Line	Rebuild line with higher-capacity conductor	\$2,994,000	100% (\$2,994,000)
Guilbeau Electric / Mount Holly 138 kV L	Rebuild line with higher-capacity conductor	\$1,164,000	100% (\$1,164,000)
S Beach / Greenback-Fairoaks 138 kV L	Rebuild line with higher-capacity conductor	\$6,255,000	100% (\$6,255,000)
Msc / Cartanza 138 kV Cir. 2	Rebuild line with higher-capacity conductor	\$3,569,000	100% (\$3,569,000)
Plainfield / Castleton 138 kV	Rebuild line with higher-capacity conductor	\$2,851,000	24% (\$682,000)
Trainer / Bridgeport 138 kV	Rebuild line with higher-capacity conductor	\$1,436,000	24% (\$345,000)
Greenback-Fairoaks / La Parguera 6603 138 kV L	Reconductor line	\$170,000	100% (\$170,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Msc / Cartanza 138 kV	Reconductor line	\$1,024,000	100% (\$1,024,000)
SuperMax / Chase 138 kV	Reconductor line	\$649,000	8% (\$51,000)
Super Max / Green Country 138 kV	Reconductor line	\$2,943,000	8% (\$230,000)
Hurricane Crk Substation / Woodlynn 138 kV	Reconductor line	\$762,000	100% (\$762,000)
Fruitland / St. Johns 138 kV	Reconductor line	\$785,000	28% (\$218,000)
St. Johns / Hathaway Street 138 kV	Reconductor line	\$344,000	28% (\$95,000)
	Total allocated to project		\$44,485,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.